

AARADHYA BAPNA

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Education

University of Waterloo

Sep 2025 – Sep 2030

Honours Mathematics and Computer Science

Waterloo, ON

- Relevant Coursework: Data Types and Structures, Calculus 3, Probability, Linear Algebra

Projects

Waterloot | *React, Next.js, TypeScript, Supabase, PostgreSQL, Tailwind CSS* | waterloot.ca Jan 2026 – Mar 2026

- Co-built and shipped a peer-to-peer marketplace at **DeltaHacks 12**, growing to **50+** active users and **20+** live listings within weeks of launch
- Built the React and Next.js frontend in TypeScript, using server-side rendering so listing pages load with content in place and stay indexable, and typed components to keep the codebase maintainable as features shipped
- Optimized listing creation from minutes to seconds by integrating the Open Library API to autofill titles, authors, and editions from an ISBN, removing the biggest drop-off point for sellers
- Enforced uwaterloo.ca-only access at the database layer with Postgres row-level security policies and Zod validation on every write, so the trust guarantee holds regardless of client behaviour

WatAsk | *Python, FastAPI, JavaScript, Supabase pgvector, Gemini API* | watask.onrender.com May 2026 – Present

- Built and shipped a full-stack application that helps students plan their degrees, covering **18** UW Math/CS majors and double-degree programs, deployed publicly and iterated on structured feedback from early student users
- Designed a RAG pipeline end to end: embedded queries with Gemini, scraped **1,600+** courses from UWFlow's API and the Waterloo/Laurier academic calendars and indexed them in Supabase pgvector, and tuned cosine-similarity search so answers cite real course reviews
- Built a course dependency graph in Python that lets students trace complex prerequisite chains, with a multi-pass traversal resolving them into a validated schedule, cross-checked against official UW degree checklists
- Deployed on Render with a CD pipeline that ships on every push to main, reducing release overhead to a single merge

Hospital Management System | *Python, SQL*

Oct 2025 – Nov 2025

- Built a records system providing full CRUD operations over patients, doctors, and appointments, replacing a manual spreadsheet workflow for school-lab staff
- Designed the relational schema and indexed queries so lookups stayed fast as record counts grew, with constraints preventing double-booked appointment slots
- Separated data access from application logic behind a dedicated query layer, using parameterized SQL statements for every read and write to keep call sites free of string-built queries

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, HTML/CSS, Racket

Frameworks & Tools: React, Next.js, FastAPI, Supabase, PostgreSQL, MySQL

AI & APIs: RAG, Vector Embeddings, Semantic Search, Supabase pgvector

Tools: Git, VS Code, Vercel, Excel

Libraries: Tailwind CSS, Zod

Extracurriculars

Treasurer | *Movie Watchers Club*

Jan 2026 – Apr 2026

- Managed a \$200 CAD term budget for a **20+** member club, tracking weekly spend on supplies and events so no movie night ran short on essentials